Activity 8: Follow-Up Action and Integration Review

Course: Fundamentals of Semiconductor Device Physics and Process Integration

Course code: TGS-2024049339

Duration: 60 minutes

TSC alignment: A5

Workplace scenario

The integration review board must disposition a yield loss involving etch overrun, dielectric stress and weak SPC reaction discipline.

Goal

Determine proportionate containment, corrective, preventive and verification actions.

Files in this folder

- README.md / README.pdf - complete procedure and acceptance criteria.
- SCENARIO.md / SCENARIO.pdf - facts, assumptions and role brief.
- WORKSHEET.md / WORKSHEET.pdf - structured learner response.
- EVIDENCE-CHECKLIST.md / EVIDENCE-CHECKLIST.pdf - completion and review record.
- Any CSV file in this folder - supplied teaching data for analysis.

Step-by-step procedure

1. Consolidate the causal chains from the earlier activities into one issue statement.
2. Rank risks by wafer exposure, defect escape, device impact and reversibility.
3. Define immediate containment for wafers, tools, recipes and downstream operations.
4. Assign corrective actions addressing recipe, integration sequence, metrology and control-plan weaknesses.
5. Specify owner, due date, evidence of completion and effectiveness metric for every action.
6. Run a peer challenge: test whether each action addresses a verified cause and whether restart criteria are objective.

Required evidence

Integration review action register with containment, CAPA, owners and effectiveness checks.

Include your completed worksheet, any calculations or annotated diagrams, the evidence checklist, and a short note identifying assumptions or unresolved evidence gaps.

Acceptance criteria

[ ] Actions are tied to verified causes

[ ] Containment and corrective action are distinct

[ ] Every action has an owner and effectiveness measure

Verification

Before submission, ask a peer to challenge one causal link. Revise the conclusion if the challenge reveals that evidence and inference have been mixed. Your final response must distinguish: observed fact, interpretation, decision, and follow-up check.

Troubleshooting

- If several mechanisms fit the symptom, choose a measurement that discriminates between them.
- If a process module lacks a defined input, output or control gate, the flow is incomplete.
- If an action is not tied to a verified cause, record it as a hypothesis test rather than corrective action.
- If calculations disagree, show the method and units so the discrepancy can be reviewed.